

1. MCQ

Choose the correct answer: Total 10 marks (0.5 mark for each question)

1. Which of the following sentences is right about COPD?
 - a. Difficult to diagnose
 - b. It occurs only in smokers
 - c. It causes airway obstruction that is not fully reversible
 - d. Can not be caused by genetic factors
2. What are the common pathogens present in COPD patients?
 - a. Streptococcus pneumonia, Hemophilus influenza
 - b. Staphylococcus pneumonia, Hemophilus influenza
 - c. Streptococcus pneumonia, acid fast bacillus
 - d. Klebsiella pneumonia
3. Which of the following best explains why HRCT is preferred over standard CT in Group 3 pulmonary hypertension?
 - A. HRCT detects pulmonary emboli better.
 - B. It identifies lung parenchymal diseases.
 - C. HRCT measures PAP directly.
 - D. HRCT has lower radiation dose.
4. ABG in severe asthma: normal/increased PaCO₂ indicates.....
 - A. Development of vocal cord dysfunction.
 - B. Development of respiratory muscles fatigue.
 - C. Development of Hyperoxia.
 - D. Development of right heart failure.
5. Gas exchange in ARDS becomes difficult primarily due to.....
 - a. Bronchial smooth muscle hypertrophy and constriction.
 - b. Hyaline membrane formation and decreased lung compliance.
 - c. Decreased airway caliber and bronchoconstriction.
 - d. Upper airway obstruction and laryngeal edema.
6. A 45-year-old, female, presented with dyspnea and dry cough. Chest X-ray showed left sided cavity with air-fluid level, patient started broad spectrum antibiotics with no response for 10 days. What is next step?
 - A. Bronchoscope for microbiological analysis of bronchoalveolar lavage.
 - B. Referral to cardiothoracic surgeon.
 - C. Thoracoscopic biopsy.
 - D. Transthoracic percutaneous needle aspiration (CT- or US-guided).
7. In patients with suspected high risk (shock and hypotension) pulmonary embolism, which of the following is the initial investigation?
 - A. CT pulmonary angiography.
 - B. Thoracic ultrasonography.
 - C. D- dimer.
 - D. Cardiac enzymes.
8. Low flow oxygen systems provide.....
 - A. Constant FiO₂.
 - B. Variable FiO₂.

- C. Fixed oxygen.
D. Pure oxygen.
9. The intensity of the aortic component of the second heart sound in patients with systemic hypertension is?
- Accentuated
 - Muffled
 - Normal
 - No change
10. The fourth heart sound is auscultated in all of the following except:
- Hypertensive patients
 - patients with atrial fibrillation
 - young adult with significant aortic stenosis
 - Hypertrophic cardiomyopathy with a sinus rhythm
11. How does the act of standing affect the venous return to the heart?
- It increases preload
 - It decreases preload
 - It has no effect on preload
 - It can either increase or decrease preload depending on the individual
12. The following investigations are considered diagnostic for pulmonary embolism except
- Computed tomography pulmonary angiography
 - Lung ventilation -perfusion scan
 - complete blood count
 - negative D-Dimer in ruling out its diagnosis
13. Which of the following is considered mandatory for diagnosis of infective endocarditis?
- Blood culture
 - chest x-ray
 - liver function test
 - thyroid function test
14. Normal PR interval duration in the Electrocardiography (ECG) ranges between
- 0.04 - 0.12 seconds (small squares of 1-3)
 - 0.12 - 0.2 seconds (small squares of 3-5)
 - 0.04 - 0.08 seconds (small squares of 1-2)
 - 0.08 - 0.12 seconds (small squares of 2-3)
15. Long-acting beta two agonists should be used with inhaled corticosteroids due to which of the following?
- Delay up regulation of receptors.
 - B-Enhance down regulation of receptors.
 - Prevent occurrence of tolerance.
 - Decrease the hypokalemic effect.
16. During the late ventricular diastole phase of the cardiac cycle, which mechanical events are occurring simultaneously?
- The atria are relaxed and the ventricles are filling via passive pressure gradients.
 - The ventricles are contracting while the semilunar valves remain firmly closed.
 - The atria contracts to finish ventricular filling while the AV valves stay open.
 - The atria are relaxed while the ventricles begin the process of rapid ejection
17. Horner's syndrome is a complication of which of the following lesion?
- Bronchial carcinoid
 - Right apical carcinoma
 - Apical lung abscess

d. Right basal carcinoma

18. What would be an appropriate first-line treatment for the management of cyanosis as a result of congenital heart disease?

- a. indomethacin administration
- b. prostaglandin E1 infusion
- c. nitric oxide
- d. surfactant

19. A 35-year-old male presents with chest pain. Imaging reveals a mass in the anterior mediastinum. Which of the following is the most likely diagnosis?

- a. Metastasis
- b. Hodgkin lymphoma
- c. Teratoma
- d. Bronchogenic cyst

20. Which inherited disease is a major cause of bronchiectasis?

- a. Alpha-1 antitrypsin deficiency
- b. Cystic fibrosis
- c. Huntington disease
- d. Sickle cell anemia

2. EMQ

Each group of extended matching questions consists of a list of numbered problems (diseases) followed by lettered options (diagnostic tests). For each numbered disease select the lettered diagnostic test that most closely diagnose the disease. You can use the lettered options once or not at all.

Total 2 marks (0.5 mark for each question)

■ Problems:

- 1. Exudative plural effusion.
- 2. DPLD.
- 3. Mediastinal mass.
- 4. Bronchogenic carcinoma

■ Options:

- A. Sputum Ziehl Nelsen stain.
- B. Diffusion lung capacity (DLCO).
- C. Mediastinoscopy.
- D. Histopathological examination of biopsy.
- E. Pulmonary function tests (PFT).
- F. Light criteria.
- G. Sputum culture and sensitivity.

3. SAQ

1. Causes of false negative tuberculin test? (1 mark, 0.25 mark for each).

a.

b.

c.

d.

2. A 25-year-old man presents in your clinic with complaining that when he is feverish especially when going to sleep, with loss of appetite. Upon taking his history, he has tooth extraction with filling of another molar tooth one month ago and history of rheumatic arthritis when he was young but he neglects the condition and not maintained on long acting penicillin. You suspect he has a heart valve problem.

a) If this man had an infective endocarditis, what are the risk factors for infective endocarditis? (1 mark) –

b) what are the investigations that confirm the diagnosis of infective endocarditis ? (1 marks)

MODEL ANSWER EOM FORM C

1	C
2	A
3	B
4	B
5	B
6	A
7	A
8	B
9	A
10	B
11	B
12	C
13	A
14	B
15	C
16	C
17	B
18	B
19	B
20	B
EMQ	
1	F
2	B
3	C
4	D